One twelfth.
Seen from where?

Vitruvius prescribes a forward inclination of one twelfth of a member’s height. Move the observer; compare the rays, the face normal and the projected spacing of equal physical marks.

Scope. Vitruvius supplies the inclination ratio. Metres, observer position, face edge length, centre rotation and the pinhole camera are chosen modern conditions. Equal ray lengths define a geometric front-on comparison here; they do not measure an ancient viewer's perception, establish a particular temple's construction, or reconstruct the absent upper building at Sardis.

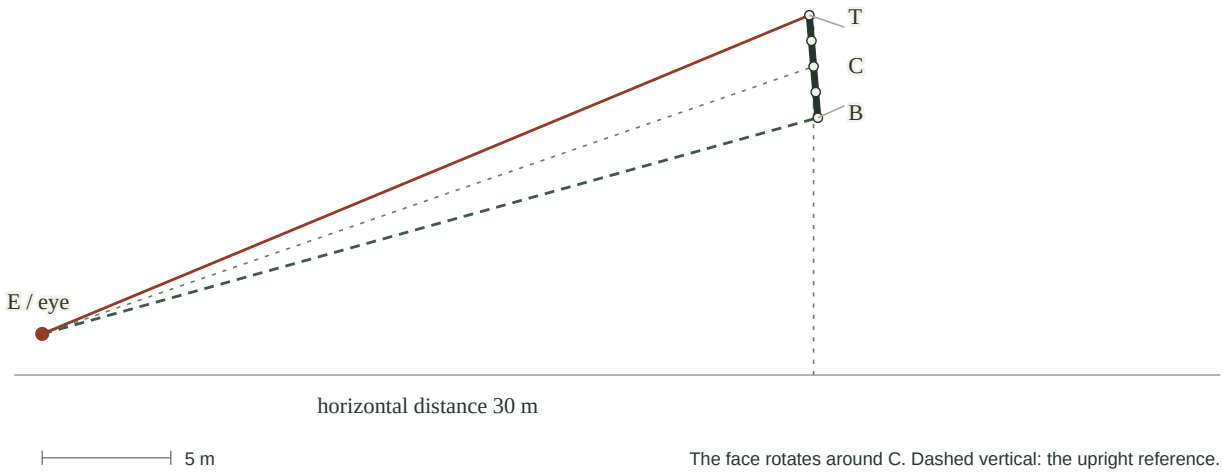
Selected treatment: **source**. Horizontal distance 30 m; face centre 12 m; eye 1.6 m; face edge 4 m. Actual forward angle 4.76364°. Saved custom angle 4.8°. Selected passage 5.

This self-contained file has no script and makes no automatic network request. Follow links to consult the online sources. New English, notes, code and diagrams: Anton Soloviev, 2026, MIT. Historical text and scans: public domain. Museum image and data: The Met Open Access, CC0.

Reopen this viewpoint: https://antonsoo.github.io/gonia/optics.html?d=300&h=120&e=16&l=40&a=48&mode=source&read=4#vp-workshop

The selected observer and member

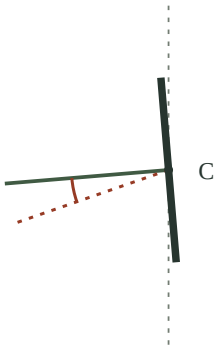
SIDE VIEW / EQUAL X–Y SCALE / CHOSEN METRES



The face rotates around C. Dashed vertical: the upright reference.

Upper ray: 32.30559 m. Lower ray: 31.31564 m. Upper minus lower: 0.98995 m. Relation: upper-longer.

DETAIL / SEPARATE SCALE, SAME ANGLES



14.356° from facing the eye

Solid green: face normal. Dotted red: direction to E.

PINHOLE SPACING / CAMERA AIMED AT C



Same face length; one quarter between physical marks.

Image scale: L / eye-to-centre distance.

Five marks divide the physical face into four equal lengths. The camera aims at C. Image heights use the face-on L/R scale, where R is the distance to C. Upper / lower projected half-length ratio: 0.96924.

Only the large side view shares one horizontal and vertical world scale. The angular detail and two normalized rulers have separate scales. No full temple, carved profile or human perceptual response is reconstructed.

Four treatments, the same observer

Treatment	Forward angle	Upper ray, m	Lower ray, m	Image half ratio
Vertical	0°	32.46167	31.15381	0.95957
Source 1:12	4.76364°	32.30559	31.31564	0.96924
Face the eye	19.11973°	31.81446	31.81446	1
Custom angle	4.8°	32.30438	31.31688	0.96932

The chosen criterion

Let d be horizontal distance, H face-centre height and e eye height. The upward tangent of a forward-leaning face is (−sin α, cos α). It is perpendicular to the centre-to-eye vector (−d, e−H) when tan α = (H−e)/d. Endpoint rays have equal lengths under the same condition.

For the 1:12 slope, d = 12(H−e). At these chosen heights the model gives 124.8 m. Vitruvius supplies no such viewing-distance rule. Equal rays are our geometric front-on criterion, not a measurement of an ancient viewer’s experience.

The source treatment’s 4 m edge has vertical rise 3.98618 m and forward offset 0.33218 m. Their ratio is 12:1. The absolute dimensions and rotation about the centre are chosen.

For a point P, let q=P−E and let v be the unit direction from the eye to C. Pinhole height is (q·v⊥)/(q·v). Multiplying by R/L gives the normalized ruler coordinate. The face-on reference has endpoints −1/2 and 1/2. This projection measures spacing, not apparent verticality.

1. Above the capitals

Vitruvius III.5.13; Krohn 1912, p. 74, lines 11–13.

Membra omnia, quae supra capitula columnarum sunt futura, id est epistylia, zophora, coronae, tympana, fastigia, acroteria

All the members which are to be above the capitals of the columns: architraves, friezes, cornices, pediment panels, gables and roof ornaments.

This is a list of upper members. The experiment isolates one chosen flat face. It does not reconstruct a whole temple or infer the geometry of every object in the list.

> supra capitula · above the capitals
>
> Supra takes the accusative. Capitula is the neuter plural of capitulum. The photograph below locates an actual Ionic capital; the upper building is absent from that museum display.

> epistylia · architraves
>
> A Latin neuter plural borrowed from Greek. The Greek headword names the beam over columns. The Greek form is a companion word for learning, not a Greek original of this Latin passage.
>
> Greek companion ἐπιστύλιον; lexicon evidence. This is a learning aid, not a Greek original of the Latin.

> acroteria · roof ornaments
>
> Another neuter plural in the list. Acroteria belong at a roof's summit or corners. The same inclination instruction follows the entire list, without giving a separate observer position for each member.

2. A twelfth part

Vitruvius III.5.13; Krohn 1912, p. 74, lines 13–14.

inclinanda sunt in fronte suae cuiusque altitudinis parte XII

They must be inclined to the front by a twelfth part of each one's own height.

The model interprets this as forward offset divided by vertical rise = 1/12. Its angle is arctan(1/12), about 4.764°. Absolute dimensions and the location of the eye are modern choices.

> inclinanda sunt · must be inclined
>
> The gerundive with sunt expresses an instruction. Inclinanda agrees with the plural membra. The text prescribes a treatment; it does not report a measured tilt for the Sardis object.

> suae cuiusque altitudinis · of each one's own height
>
> The genitive phrase identifies the height whose fraction is taken. The model preserves a chosen face edge length while rotating around its centre, so both its vertical rise and its forward offset are shown explicitly.

> parte XII · by a twelfth part
>
> An ablative of measure. XII is twelve; the phrase gives one twelfth of the height. The degree readout is our conversion of that ratio.

3. Two rays from the eye

Vitruvius III.5.13; Krohn 1912, p. 74, lines 14–16.

ideo quod, cum steterimus contra frontes, ab oculo lineae duae si extensae fuerint et una tetigerit imam operis partem, altera summam, quae summam tetigerit, longior fiet

For this reason: when we stand facing the fronts, if two lines are extended from the eye, one touching the work's lowest part and the other its highest, the one touching the highest will be longer.

The sentence introduces the author's explanation. In this experiment the endpoints belong to one chosen member. Change the observer and measure both Euclidean ray lengths; the text supplies no viewing distance.

> ab oculo · from the eye
>
> Ab takes the ablative oculo. The eye is the origin of both drawn rays. Its selected height is in modern metres, not a recovered ancient observer height.

> una … altera · one … the other
>
> The paired feminine forms refer to the two lineae. In the diagram, the upper ray is solid vermilion and the lower ray is dashed green; the distinction does not depend on colour alone.

> longior fiet · will become longer
>
> Longior is the comparative agreeing with the line. Fiet is the future of fio. The readout compares the unrounded lengths, including cases where the upper ray becomes shorter after a chosen forward tilt.

4. An appearance of leaning back

Vitruvius III.5.13; Krohn 1912, p. 74, lines 17–18.

ita quo longior visus linea in superiorem partem procedit, resupinatam facit eius speciem

Thus, as the line of sight extends farther to the upper part, it makes that part's appearance seem to lean backward.

This is a claim about appearance. Our pinhole diagram measures projected spacing and a face-normal angle. Those geometric quantities do not establish what an ancient observer actually perceived.

> visus linea · line of sight
>
> Visus is genitive here: a line of vision. The camera comparison aims along the line from the chosen eye to the face's centre C. Both physical rays and that aiming line stay visible.

> resupinatam · leaning backward
>
> The participle describes speciem, the appearance. It is not a statement that the drawn stone has physically rotated backward. Compare the unchanged physical divisions with their projected spacing.

> speciem · appearance
>
> Species can concern how something presents itself to sight. The experiment labels its outputs as geometry rather than replacing this historical perceptual claim with a modern measurement.

5. Plumb and square

Vitruvius III.5.13; Krohn 1912, p. 74, lines 18–20.

cum autem, uti supra scriptum est, in fronte inclinata fuerint, tunc in aspectu videbuntur esse ad perpendiculum et normam

But when they have been inclined to the front, as written above, then in view they will seem to be plumb and square.

Equal endpoint ray lengths are our deliberately chosen front-on criterion. A fixed 1:12 slope meets it at distance = 12 × (face-centre height minus eye height). That relation follows from this model, not from a viewing-distance rule stated by Vitruvius.

> videbuntur esse · will seem to be
>
> The future passive of video with esse expresses how the members will appear. Keep that wording beside the model's more limited, numerical front-on criterion.

> ad perpendiculum · to a plumb line
>
> Perpendiculum is a plumb line. The Greek companion κάθετος can name a perpendicular or plumb line. A physical vertical, a face normal and a line of sight are three distinct directions in the drawing.
>
> Greek companion κάθετος; lexicon evidence. This is a learning aid, not a Greek original of the Latin.

> normam · a builder's square
>
> Norma names the square or standard of rightness. In the modern comparison, a face points at the eye when its normal aligns with the aiming line. That does not turn the image into a psychological test.

6. A reasoned adjustment

Vitruvius III.5.9; Krohn 1912, p. 73, lines 3–4.

quare semper adiciendum est rationis supplementum in symmetriarum membris

Therefore a reasoned allowance must always be added in the proportioned members.

The preceding passage discusses sight at height, including an ancient explanation involving the density of air. That theory is not used by the pinhole model. This short clause is context for the author's adjustments, not a numerical law for all buildings.

> adiciendum est · must be added
>
> Another gerundive instruction, singular here with supplementum. Compare it with the plural inclinanda sunt in the later prescription.

> rationis supplementum · a reasoned allowance
>
> The wording calls for an adjustment. It does not supply this experiment's camera, eye height, face length or rotation centre. Those choices are retained with every exported study.

> symmetriarum membris · in the proportioned members
>
> The context is architectural proportion. The metre values are chosen examples; the 1:12 inclination is the specific source quantity explored here.

Locate a capital, not a measured tilt



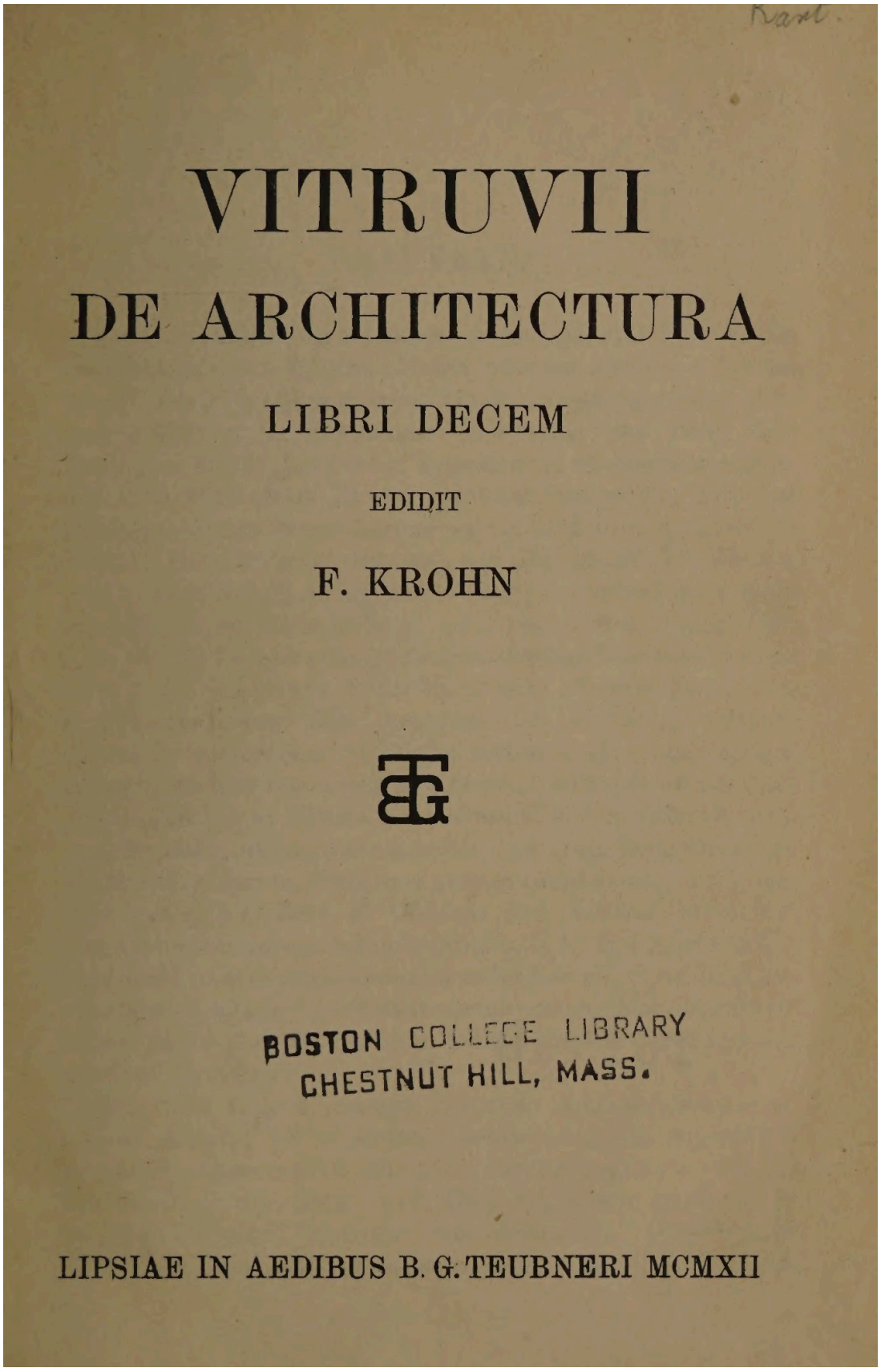
Marble column from the Temple of Artemis at Sardis, Greek, ca. 300 BCE. The Met, 26.59.1. Gift of The American Society for the Excavation of Sardis, 1926. CC0; original JPEG unchanged.

The volutes locate the capital. The source's upper members belong above it; the upper building is absent. The display is reassembled from one or more columns; parts of the shaft are restored and the profiled base below the torus is a copy. The modern museum columns behind it are separate objects.

The museum's object record · The Met Open Access policy. This photograph is not a source for the model's inclination.

Title · Krohn 1912

The 1912 title page. PDF page 5. Complete page, including apparatus where present.



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Printed page 72 · Krohn 1912

Sight at height: beginning of III.5.9. PDF page 90. Complete page, including apparatus where present.

ant proiecturam, uti circini centrum unum cum sit positum in capituli tetrante et alterum diducatur ad extremum cymatium, circumactum balteorum extremas partes tangat. axes volutarum ne crassiores sint quam oculi magnitudo, volutaeque ipsae succidantur altitudinis suae duodecimam partem. (20) haec erunt symmetriae capitulorum, quae columnae futurae sunt ab minimo ad pedes XXV. quae supra erunt, reliqua habebunt ad eundem modum symmetrias, abacus autem erit longus et latus, quam crassa columna est ima adiecta parte VIIII, uti, quo minus habuerit altior columna contractum, 80 eo ne minus habeat capitulum suae symmetriae proiecturam 8 et in latitudine suae partis adiectionem. de volutarum descriptionibus, uti ad circinum sint recte involutae, quemadmodum describantur, in extremo libro forma et ratio earum (5) erit subscripta.

Capitulis perfectis, deinde ⟨in scapis⟩ columnarum non ad libellam sed ad aequalem modulum conlocatis, ut, quae adiectio in stylobatis facta fuerit, in superioribus membris respondeat [symmetria epistyliorum], epistyliorum ratio sic est habenda, uti, si columnae fuerint a minima XII pedum (10) ad quindecim pedes, epistylii sit altitudo dimidia crassitudinis imae columnae; item ⟨si⟩ ab XV pedibus ad XX, columnae altitudo dimetiatur in partes tredecim, et unius partis altitudo epistylii fiat; item si a XX ad XXV pedes, dividatur altitudo in partes XII et semissem, et eius una pars (15) epistylium in altitudine fiat; item si ab XXV pedibus ad XXX, dividatur in partes XII, et eius una pars altitudo fiat. item ⟨si quae altiores erunt,⟩ rata parte ad eundem modum ex altitudine columnarum expediendae sunt altitudines epi- 9 styliorum. quo altius enim scandit oculi species, non facile

2 diducatur *G*: deducatur *HS* 4 ne *Joc*: nec *x* 5 succidantur *Mar*: sic cedantur *x* 6 capitulorumque *HS* 12 latitudine *rec*: altitudine *x* suae partis H^cSG^c: repartis *HG* 16 *add. Joc* 19 *del. Meister* 22 *add. Phil* quindecī *S* 23 demetiatur *x* 24 viginti *S* 25 *et* 27 duodecī *S* 28 *add. Ro* (*sic*: si quae supra altiores fuerint: *cf. 74, 22*) rata parte *Schn* (pro rata parte *Mar*): ratā partē (-tem *H*) *x*

Consult the identified original volume.

Printed page 73 · Krohn 1912

A reasoned allowance and upper-member proportions. PDF page 91. Complete page, including apparatus where present.

persecat aeris crebritatem; dilapsa itaque altitudinis spatio et viribus exuta incertam modulorum renuntiat sensibus quantitatem. quare semper adiciendum est rationis supplementum in symmetriarum membris, ut, cum fuerint aut altioribus locis opera aut etiam ipsa colossicotera, habeant magnitudinum rationem. epistylii latitudo in imo, quod supra capitulum erit, quanta crassitudo summae columnae sub capitulo erit, tanta fiat; summum, quantum imus scapus. cymatium epistylii septima parte suae altitudinis est faciendum, et in proiectura tantundem. reliqua pars praeter cymatium dividenda est in partes XII, et earum trium ima fascia est facienda, secunda IIII, summa V. item zophorus supra epistylium quarta parte minus ⟨altus⟩ quam epistylium; sin autem sigilla designari oportuerit, quarta parte altior quam epistylium, uti auctoritatem habeant scalpturae. cymatium suae altitudinis partis septimae; proiecturae cymatiûm quantum crassitudo. supra zophorum denticulus est faciendus tam altus quam epistylii media fascia; proiectura eius quantum altitudo. intersectio, quae graece μετοπη dicitur, sic est dividenda, uti denticulus altitudinis suae dimidiam partem habeat in fronte, cavus autem intersectionis huius frontis e tribus duas partes; huius cymatium altitudinis eius sextam partem. corona cum suo cymatio, praeter simam, quantum media fascia epistylii; proiectura coronae cum denticulo facienda est, quantum erit altitudo a zophoro ad summum coronae cymatium; et omnino omnes ecphorae venustiorem habeant speciem, quae quantum altitudinis tantundem habent proiecturae. tym-

2 exuta *Lor*: extructam *H*, exstructā *S*, exstructa *G* 3 rationis *G*: rationi *HS* 4 ut cum fuerint aut *Joc*: cum fuerint ut aut *x* 5 colossicotera *Joc*: colossi caetera *HS*, colosi cetera *G* 9 cymatium epistylii *ed*: cimatiū epistiliū *G*, cymatii epystilii (epist. *S*) *HS*, cymatiu epistilii S^c 11 ima *Polenus*: iam *x* 13 *add. Kr* 15 altior *Phil*: altiorē (-em) *x* 17 quantum *Mar*: quam *x* 18 fascia *Joc*: fastigia *x* 20 metoce *HS*, metache *G* (metoche G^c) 21 intersectionis *G*: -nes *HS* 22 et tribus *GS* 28 altitudinis GS^c: -nes *HS* habent *Kr*: habeant *x*

Consult the identified original volume.

Printed page 74 · Krohn 1912

The 1:12 inclination and its sight-line explanation. PDF page 92. Complete page, including apparatus where present.

pani autem, quod est in fastigio, altitudo sic est facienda,
uti frons coronae ab extremis cymatiis tota dimetiatur in 82
partes novem et ex eis una pars in medio cacumini tym-
pani constituatur, dum contra epistylia columnarumque
hypotrachelia ad perpendiculum respondeant. coronae- 5
que supra aequaliter imis praeter simas sunt conlocandae. (5)
insuper coronas simae, quas Graeci επαιετιδας dicunt, facien-
dae sunt altiores octava parte coronarum altitudinis. acro-
teria angularia tam alta, quantum est tympanum medium,
mediana altiora octava parte quam angularia. 10
13 Membra omnia, quae supra capitula columnarum sunt (10)
futura, id est epistylia, zophora, coronae, tympana, fastigia,
acroteria, inclinanda sunt in fronte suae cuiusque altitudinis
parte XII, ideo quod, cum steterimus contra frontes, ab oculo
lineae duae si extensae fuerint et una tetigerit imam operis 15
partem, altera summam, quae summam tetigerit, longior fiet. (15)
ita quo longior visus linea in superiorem partem procedit,
resupinatam facit eius speciem. cum autem, uti supra scrip-
tum est, in fronte inclinata fuerint, tunc in aspectu vide-
14 buntur esse ad perpendiculum et normam. columnarum 20
striae faciendae sunt XXIIII ita excavatae, uti norma in cavo (20)
striae cum fuerit coniecta, circumacta anconibus ⟨angulos⟩
striarum dextra ac sinistra tangat cacumenque normae cir-
cum rotundationem tangendo pervagari possit. crassitudines
striarum faciendae sunt, quantum adiectio in media columna 25
15 ex descriptione invenietur. in simis, quae supra coronam (25)
in lateribus sunt aedium, capita leonina sunt scalpenda,
disposita ⟨ita⟩, uti contra columnas singulas primum sint 83
designata, cetera aequali modo disposita, uti singula singulis
mediis tegulis respondeant. haec autem, quae erunt contra 30

---

2 ut *S* 3 cacumini *Kr*: -ne *x* 4 epistylia *Joc*: -lii *x* 7 epitidas *HG*, epytidas *S* (*corr. Böttticher*) 12 coronae *Joc*: corona & *x* 13 fronte *S*: frontis HG^c 16 alterā *S et a. ras. H* 17 procedet *a. corr. G* 18 resopinatam *S et a. corr. H* 19 fuerint *Joc*: fuerit *x* 22 *add. Joc* (*post* sinistra) 23 cacumen-que *Ussing*: acumen quae *x* 24 rotundationem *rec*: -ne *x* crassitudinis *H* 28 *add. Ro* 30 autem *om. S*

Consult the identified original volume.

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Vitruvii De architectura libri decem, ed. F. Krohn, Leipzig: B. G. Teubner, 1912. The Latin excerpts join line-end hyphens and capitalize XII. They are selected clauses, not complete chapters. The preceding discussion in III.5.9 includes an ancient air-density explanation; our geometric analysis does not use it.

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GONIA: ONE TWELFTH, SEEN FROM WHERE?

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MUSEUM OBJECT AND PHOTOGRAPH
Marble column from the Temple of Artemis at Sardis, Greek, Hellenistic,
ca. 300 BCE. Metropolitan Museum of Art, object 252453, accession 26.59.1.
Gift of The American Society for the Excavation of Sardis, 1926.
https://www.metmuseum.org/art/collection/search/252453
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The original 3738 x 3790 JPEG is retained unchanged. The displayed ancient
fragments were reassembled from one or more columns; parts of the shaft are
restored and the profiled base below the torus is a copy. The modern museum
columns in the background are not the object under discussion. This photograph
locates the capital; it supplies no measured tilt for the upper building.

TEACHING TEXT, CODE AND DIAGRAMS
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English translations, contextual notes, diagrams and the pinhole analysis are
new. Ancient Greek companion words are learning aids, not a Greek original of
Vitruvius. In particular, epistylia is connected with the Greek headword
epistylion, as recorded in the Perseus/Scaife lexicon linked in the reading.

The text supplies the 1:12 inclination. Modern metres, observer position,
fixed face length, rotation about its centre, the camera aimed at that centre,
and equal-mark comparisons are explicitly chosen model conditions. They are
not ancient measured dimensions or a psychophysical experiment.

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Code, new teaching text and diagrams


```
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Sardis","culture":"Greek","period":"Hellenistic","dynasty":"","reign":"","portfolio":"","artistRole":"","artistPrefix":"","artistDisplayName":"","artistDisplayBio":"","artistSuffix":"","artistAlphaSort":"","artistNationality":"","artistBeginDate":"","artistEndDate":"","artistGender":"","artistWikidata_URL":"","artistULAN_URL":"","objectDate":"ca. 300 BCE","objectBeginDate":-300,"objectEndDate":-280,"medium":"Marble","dimensions":"H. 142 1/8 in. (361 cm)","measurements":
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